Fourth Semester B. Tech. (Electronics and Communication Engineering) Examination

MICROPROCESSORS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions carry marks as indicated.
(2) Due credit will be given to neatness and diagrams.
(3) Use of electronic gadgets leaving scientific calculator is not allowed.
(4) Assume suitable data wherever necessary.

1. (a) Recall Pins of 8085 microprocessor and explain the significance of following pins in brief –

   (a) HLDA

   (b) IO / $\overline{\text{M}}$

   (c) X1, X2

   (d) INTR

   (e) ALE 5(CO1)

   (b) Explain interrupts in 8085 with neat interrupt structure Diagram. 5(CO3)

2. (a) Using an assembly language program to demonstrate division operation in 8085. Store the result in quotient and remainder format.

   (Assume source address as 2000H and Destination address as A000H and A001H respectively) 5(CO2)

   (b) Deconstruct instruction MVI A, 07h with neat Timing diagram / Waveform and explain the same. 5(CO1,3)

3. (a) Integrate 8 led's with 8255 at port B and 8085 where chip select is at 80h. Draw interfacing diagram for the same where led's in common anode configuration. Write a program for the same configuration to demonstrate alternate blinking led pattern. 10(CO2,3)

4. (a) Retrieve with a neat diagram and explain the flag structure of 8086. 5(CO1)

   (b) Attribute what you understand by minimum mode in 8086. 5(CO4)

5. (a) Explain the register set of 8086 Microprocessor. 5(CO3)

   (b) Using example instructions explain the working and need of stack memory in 8086 microprocessor. 5(CO2,3)

6. (a) Deconstruct 8086 Assembly Language program to arrange 10 bytes of data in ascending order. 5(CO2)

   (b) Recall and explain the important features of Intel 286 microprocessor. 5(CO4)